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Fundamental of Database

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Background

TOTAL ENERGIES

TotalEnergies (commonly referred to as Total) is a broad energy company that produces and markets fuels, natural gas, and electricity. TotalEnergies has been operating in Ethiopia since 1950. It began as a branch of the French "Compagnie Française des Pétroles" (CFP) and has since transformed into a major local entity. In 2021, following the global rebranding of the parent company, it officially changed its name to TotalEnergies Marketing Ethiopia.

Infrastructure: The company established its first major depot in Akaki in the 1950s. Today, it operates modern logistics hubs, including a state-of-the-art fuel and LPG depot in Dukem (commissioned in 2016) and multiple aviation depots at major.

- **Market Position:**

TotalEnergies is a top-tier retailer in Ethiopia's energy sector, operating a network of over 143 service stations nationwide.

- **Core Products:**

Fuels: Petrol (Benzene), Diesel (AGO), and Kerosene.

- **Lubricants:**

High-quality engine oils and car care products.

- **LPG:**

Cooking gas for domestic and industrial use.

- **Renewables:**

Recently, the company inaugurated its first Fast Electric Vehicle (EV) Charging station in Ethiopia, signaling a shift toward sustainable energy.

Mission and Vision (Ethiopian Context)

Mission: To provide safe, clean, and affordable energy while supporting Ethiopia's economic growth through reliable supply chains and community initiatives (such as the "Startupper of the Year" challenge).

Vision: To remain the preferred energy retailer in Ethiopia by transforming traditional service stations into multi-service hubs (Bonjour shops, car wash, and cafés).

Operation of TOTAL energies

In Ethiopia today, the fuel crisis—exacerbated by global supply disruptions in the Middle East and the closure of the Strait of Hormuz—is being managed through a high-stakes combination of strict prioritization and digital oversight.

The "operation system" is no longer just about selling fuel; it is now a triage system designed to keep the national economy from stalling. Here is how it works:

Because supply has dropped significantly (diesel supply was reportedly cut by nearly 50% recently), the system now automatically prioritizes specific "essential" sectors:

Tier 1: Public mass transit (buses), ambulances, and vehicles transporting essential food and goods.

Tier 2: Agricultural machinery (tractors) and major export-oriented projects.

Tier 3: Private vehicles. In Addis Ababa, this often results in queues lasting several days.

Requirement analysis

unreliable, causing people to arrive late to work, miss classes, or struggle to carry out their responsibilities. What appears to be a simple queue at a fuel station therefore affects the daily life and productivity of a large portion of society.

Furthermore, these queues often stretch onto main roads, creating traffic congestion and disrupting the normal flow of transportation. The lack of a structured system leads to confusion and unfairness, as there is no clear way to determine who should be served first. During fuel shortages, the situation becomes even worse. Fuel stations are currently facing serious operational challenges due to the lack of a proper database system to manage vehicle queues and fuel distribution. In Ethiopia, drivers are often forced to spend many hours waiting in long physical queues, sometimes for an entire day, without any certainty of when they will be served. This situation creates not only frustration but also a significant loss of productive time. For public taxi drivers, this directly affects their daily income, as time spent waiting in line means fewer trips and reduced earnings.

The impact of this problem extends beyond the drivers themselves. Government workers, students, and many other citizens depend on public transportation for their daily activities. When

taxis are stuck in fuel queues, transportation becomes limited and more difficult, as there is no mechanism to control fuel distribution, allowing some vehicles to take more than their fair share while others leave empty.

Improving both efficiency and fairness in fuel distribution. The system calculates an estimated waiting time for each driver and sends real-time notifications about their queue status. The queue is strictly managed on a first-come, first-served basis, while giving priority to public transportation vehicles to ensure essential services are not disrupted.

The system manages station capacity by dynamically tracking fuel availability. During registration, the specific fuel volume requested by a driver is subtracted from the station's total available stock; once the stock is depleted, the system stops accepting registrations for that day. To ensure public safety, a dedicated portion of fuel is held in reserve for emergency vehicles, which are granted immediate priority and access to this reserved supply.

Furthermore, the system enforces a strict arrival policy: if a vehicle is not available when its turn is called, the system automatically bypasses it and proceeds to the next car in the queue to maintain flow. Simultaneously, a 20-minute grace period begins for the missing vehicle. If the driver fails to appear within this 20-minute window, the system automatically erases the car's ID from the registration list, and that fuel quota is immediately released to become available for other drivers.

Conceptual design

1. Entity: STATION

Manages the physical station and the current fuel supply.

- Station_ID (PK): Atomic.
- Station_Name: Atomic.
- Location: Composite (City, Sub-city, Wereda).
- Daily_Quota: Composite (Benzene_Stock, Diesel_Stock)

2. Entity: WORKER

- Worker_ID (PK): Atomic.
- Role: Atomic (Manager, Attendant).
- Station_ID(FK)

3. Entity: VEHICLE

Tracks car details and enforces fuel limits based on the car's purpose.

- Plate_Number (PK): Atomic.
- Service_type: Atomic. Code 1-public transport, Code 2- private, Code 3-Heavy Truck.
- Max_Allowed_Fuel: Atomic. (New Rule: The maximum litres this car is allowed to buy per day—55L for Public transport, 35L for Private, 500L for Heavy truck).
- Active_Booking_Lock: Atomic. (Boolean: Prevents multiple registrations).
- Fuel_Type_Requirement: Simple (Benzene or Diesel).

4. Entity: RESERVATION (The Digital Queue)

Handles the digital queue logic, priority sorting, and timed verification.

- Reservation_ID (PK): Atomic.
- Fuel_Amount_Requested: Atomic. (Constraint: This value cannot exceed the Max_Allowed_Fuel for the vehicle).
- OTP_Code: Atomic. (Verification code for the pump).
- Queue_Status: Atomic. (Waiting, Fulfilled, Expired).
- Estimated_Wait_Time: Derived. (Calculated based on Reservation_Time + Queue_Position*6min + 20 minutes. If the current time exceeds this, the code becomes invalid).
- Queue_Position: Atomic.
- Plate_Number (FK)
- Station_ID(FK)

5. Entity: MONEY_TRANSACTION

The final record of the fuel purchase and sale.

- Transaction_ID (PK): Atomic.
- Unit_Price: Atomic.
- Total_Price: Derived. (Calculated as Amount_Dispensed $\times$ Unit_Price).
- Date_Time: Composite (Date, Time).
- Reservation_ID (FK): Atomic.
- Verifier_Worker(FK): Atomic

6. Entity: CUSTOMER

- Customer_ID (Primary Key): Simple, Atomic.
- Full_Name: Composite (First_Name, Last_Name).

- Phone_Number: Single-valued, Atomic.
- Username_Password(hashed): Composite(Username and Password)
- Plate-Number(FK)

Relationship Analysis and Relationship Constraints

1. STATION and WORKER

STATION (Total) $\longleftrightarrow$ (Total) WORKER [Cardinality 1:N]

The WORKER **WORKS_ON** STATION relationship is one-to-many. Each STATION must employ at least one worker to be functional, representing Total Participation for the STATION. Each employee is restricted to working at exactly one STATION at a time. Resulting in the total participation of (1,1).

The worker table contains Station_ID as a Foreign Key.

2. CUSTOMER and VEHICLE

CUSTOMER (Total) $\longleftrightarrow$ (Total) VEHICLE [Cardinality 1: 1]

The VEHICLE **OWNED_BY** CUSTOMER relationship is a one-to-one (1:1) mapping. Both entities have Total Participation (1, N), as a car must have an owner to be in the system, and a CUSTOMER must register at least one vehicle to participate in the fuel queue.

The CUSTOMER table contains Plate_number as a Foreign Key.

3. STATION and RESERVATION

STATION (Partial) $\longleftrightarrow$ (Total) RESERVATION [Cardinality 1:N]

A STATION **manages** multiple digital queue slots (RESERVATION), establishing a 1:N relationship with RESERVATION. Every reservation record must be associated with a specific geographic station to be valid (Total Participation). Conversely, a station might have an empty queue during periods of no fuel shortage, leading to Partial Participation for the STATION.

The RESERVATION table contains Station_ID as a Foreign Key.

4. VEHICLE and RESERVATION

VEHICLE (Partial) <—> (Total) RESERVATION [Cardinality N:1]

This 'VEHICLE **REGISTERED** RESERVATION' relationship is designed to enforce system integrity. A single RESERVATION must be linked to exactly one VEHICLE. While a vehicle exists in the database even when it is not in a queue (Partial Participation 0,1), once a reservation is made, it is locked to that specific plate number to prevent "double-booking" or selling queue spots.

The RESERVATION table contains Plate_Number as a Foreign Key.

5. WORKER and MONEY_TRANSACTION

WORKER (Partial) <—> (Total) MONEY_TRANSACTION [Cardinality 1:N]

This 'WORKER **CONTROLS** MONEY_TRANSACTION' relationship ensures accountability. Each MONEY_TRANSACTION must be processed by exactly one station worker (Total Participation 1,N). However, a WORKER may not have performed any transactions yet on a given shift, resulting in Partial Participation (0,1) for the WORKER.

The MONEY_TRANSACTION table contains Verifier_Worker as a Foreign Key.

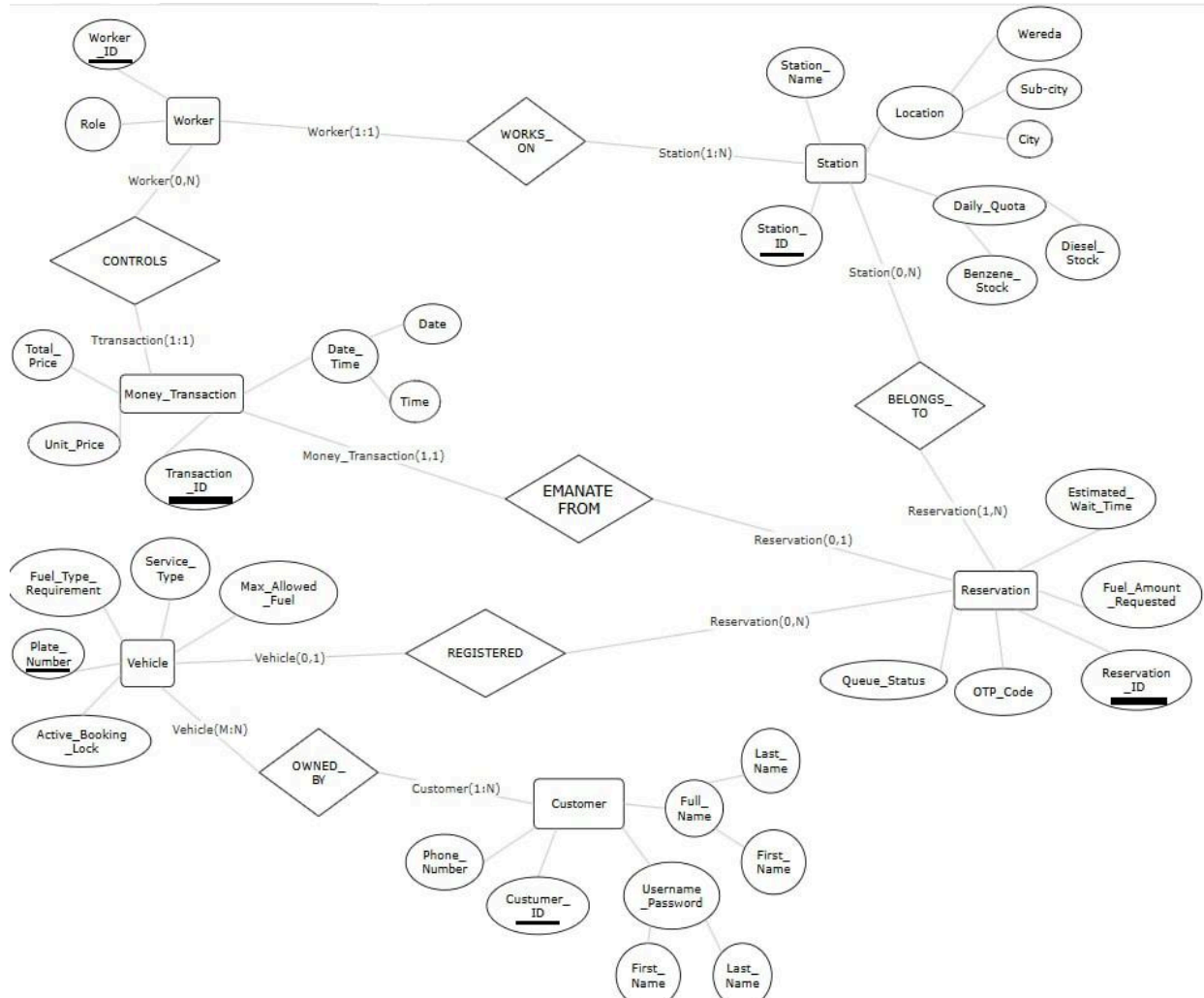
6. RESERVATION and MONEY_TRANSACTION

RESERVATION (Partial) <—> (Total) MONEY_TRANSACTION [Cardinality 1:1]

The 'RESERVATION **EMANAT FROM** MONEY_TRANSACTION' is linked in a 1:1 relationship. A transaction cannot exist without a verified prior reservation (Total Participation 1,1). However, a reservation might expire or be cancelled without ever resulting in a sale, making the participation of the Reservation entity Partial (0,1).

The MONEY_TRANSACTION table contains Reservation_ID as a Foreign Key.

ER-Diagram



Mapping



Conclusion

In conclusion, the proposed database management system for TotalEnergies Marketing Ethiopia provides a strategic digital solution to the critical challenges of fuel distribution and inefficient queue management. By replacing unpredictable physical lines with a structured digital reservation system, the database ensures a fair, transparent, and prioritized distribution process that supports essential sectors like public transportation and emergency services. Through real-time inventory tracking and automated arrival policies, the system not only optimizes station operations and reduces traffic congestion but also minimizes the socioeconomic impact of fuel shortages on the community. Ultimately, this project demonstrates how the application of relational database principles can modernize energy retail, enhancing both operational efficiency for TotalEnergies and daily productivity for the citizens of Ethiopia.